

JESSE CHEN

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PhD Student ✦ University of Arizona ✦ Tucson, AZ

EDUCATION

University of Arizona

PhD in Computer Science

Expected May 2027

Master's of Science in Mechanical Engineering

Dec 2023

Bachelor's of Science in Computer Science & Mechanical Engineering

Jan 2017 - May 2022

Minor in Engineering Management

GPA: 3.50+/4.00

PUBLICATIONS (UNDERLINE INDICATES FIRST AUTHOR)

1. [IEEE S&P 26] Jesse Chen, Rubin Yuchan Yang, Ahmad Musa, Syed Rafiul Hussain, Omar Haider Chowdhury, Sazzadur Rahaman. *Secret State Leakage Attacks and their Impacts on EMV Contactless Payment Apps*. IEEE Symposium on Security and Privacy. 2026.
2. [FSE 25] Tanner Finken, Jesse Chen, Sazzadur Rahaman. *On the Characteristics and Impacts of Protestware Libraries*. Proceedings of the ACM on Software Engineering (FSE). 2025.
3. [FSE 24] Jesse Chen, Dharun Anandayuvraj, James C. Davis, Sazzadur Rahaman. *On the Contents and Utility of IoT Cybersecurity Guidelines*. Proceedings of the ACM on Software Engineering (FSE). 2024.

WORK EXPERIENCE

Graduate Research Assistant

Aug 2022 - present

Department of Computer Science, Security - Privacy - Reliability Lab

Tucson, AZ

- Analysis of the Contactless Payment Protocol (Python, Java, JEB)

- Formally verified the EMV contactless protocol to detect attacks on security properties using the Tamarin cryptographic prover
- Developed an static and dynamic analysis tool (EMVResilienceChecker) that detects the usage of 7 Android app protection techniques in payment apps: TEE usage, root checking network integrity, anti-repackaging, anti-hooking, anti-debugging, and code obfuscation
- Reversed engineered contactless payment Android apps to extract secret data elements, demonstrating a controlled attack on a mobile payment application in an isolated environment

- Protestware Libraries (Python)

- Collected 163 protestware using Internet search and GitHub API
- Used OpenAI and Gemini API to filter 4,000 READMEs containing protesting content from 900,000 READMEs
- Collected and qualitatively analyzed 97 sentiment comments using Reddit API

- IoT Security Guidelines (Python, JavaScript, HTML, CSS)

- Created a novel taxonomy of IoT security recommendations from 25 guidelines
- Visualized taxonomy with GoJS into a tree with multiple features and data encodings

Graduate TA

Department of Computer Science

Tucson, AZ

- CSC 345 - Analysis of Discrete Structures (Java)

Spring 2022

- CSC 337 - Web Development (MERN stack)

Fall 2023, Spring 2025

- CSC 110 - Intro to Computer Programming I (Python)

Summer 2025

- CSC 352 - Systems Programming and Unix (C)

Spring 2026

- CSC 465 - Reverse Engineering vs. Software Protection (Ghidra, x86, C)

Spring 2026

Graduate RA Software Developer (JavaScript, React-Redux, AWS)

University of Arizona Administration CatCoin Project

- Implemented shopping cart frontend using React and backend using AWS DynamoDB

Oct 2022 - Jan 2023

Tucson, AZ

AWARDS

Outstanding Poster Award, Department of Computer Science, University of Arizona

2026

Excellence Award, College of Science, University of Arizona

2026

Graduate Scholarship Award, Department of Computer Science, University of Arizona

2026

GRANTS

Student Travel Grant, IEEE Symposium on Security and Privacy (\$1500)

2026

Travel Grant University of Arizona Student Travel Grant, ACM SIGSOFT CAPS (\$500)

2024

Department of Computer Science Fellowship, University of Arizona (\$5,000)

2023

Student Travel Grant, IEEE Symposium on Security and Privacy (registration waiver)

2021

SKILLS

Languages

Python, Java, JavaScript, HTML, CSS, C, C++, SQL

Technologies

Tamarin, Ghidra, JEB, MongoDB, Node.js

Natural Languages

English (fluent), Mandarin (fluent)